# Special Relativity Notes

Covariant Derivative, Christoffel Symbols and Vector Calculus in Curvilinear Coordinates

30/10/26

**Reading guide.** Yellow boxes mark material the lecture left as an exercise, or where the board only showed the final answer; the full derivations have been written out there. Everything outside the yellow boxes follows the notes.

## 1 Covariant Derivative and the Gradient ($\vec{\nabla}$)

$$\vec{V} = V^k \, \vec{\epsilon}_k \qquad (\vec{\epsilon}_k \to \text{base vector})$$

**Gradient**

$$\vec{\nabla}\psi = \frac{\partial \psi}{\partial q^j} \, \vec{\epsilon}^{\,j} \qquad \Longrightarrow \qquad \vec{\nabla} = \vec{\epsilon}^{\,j} \, \frac{\partial}{\partial q^j}$$

The derivative of the vector is written with a semicolon (the **covariant derivative**):

$$\frac{\partial \vec{V}}{\partial q^k} = V^j{}_{;k} \, \vec{\epsilon}_j$$

Expanding with the product rule:

$$\begin{aligned} \frac{\partial \vec{V}}{\partial q^k} = \frac{\partial}{\partial q^k}\left(V^j \vec{\epsilon}_j\right) &= \frac{\partial V^j}{\partial q^k} \, \vec{\epsilon}_j + V^j \, \frac{\partial \vec{\epsilon}_j}{\partial q^k} \\ &= \frac{\partial V^j}{\partial q^k} \, \vec{\epsilon}_j + V^j \, \Gamma^\mu{}_{jk} \, \vec{\epsilon}_\mu \qquad (\Gamma^\mu{}_{jk} : \ \text{Christoffel symbol}) \end{aligned}$$

The dummy indices in the last term are $j$ and $\mu$; interchange $j \leftrightarrow \mu$:

$$\frac{\partial \vec{V}}{\partial q^k} = \frac{\partial V^j}{\partial q^k} \, \vec{\epsilon}_j + V^\mu \, \Gamma^j{}_{\mu k} \, \vec{\epsilon}_j = \left( \frac{\partial V^j}{\partial q^k} + \Gamma^j{}_{\mu k} V^\mu \right) \vec{\epsilon}_j = V^j{}_{;k} \, \vec{\epsilon}_j$$

**Covariant derivative**

$$\boxed{V^j{}_{;k} = \frac{\partial V^j}{\partial q^k} + \Gamma^j{}_{\mu k} \, V^\mu}$$

## 2 Christoffel Symbols of the First and Second Kind

$\Gamma^{\mu}{}_{jk}$ is the Christoffel symbol, defined by

$$\frac{\partial \vec{\epsilon}_j}{\partial q^k} = \Gamma^{\mu}{}_{jk}\, \vec{\epsilon}_\mu$$

### Second kind

Dot with $\vec{\epsilon}^{\,m}$ and use $\vec{\epsilon}^{\,m} \cdot \vec{\epsilon}_\mu = \delta^m_\mu$:

$$\vec{\epsilon}^{\,m} \cdot \frac{\partial \vec{\epsilon}_j}{\partial q^k} = \vec{\epsilon}^{\,m} \cdot \Gamma^{\mu}{}_{jk}\, \vec{\epsilon}_\mu = \Gamma^{\mu}{}_{jk}\, \delta^m_\mu$$

**Definition: Christoffel symbol of the 2nd kind**

$$\vec{\epsilon}^{\,m} \cdot \frac{\partial \vec{\epsilon}_j}{\partial q^k} = \Gamma^{m}{}_{jk}$$

### First kind

Since $g^{mn}\vec{\epsilon}_n = \vec{\epsilon}^{\,m}$,

$$g^{mn}\, \vec{\epsilon}_n \cdot \frac{\partial \vec{\epsilon}_j}{\partial q^k} = \vec{\epsilon}^{\,m} \cdot \frac{\partial \vec{\epsilon}_j}{\partial q^k} = \Gamma^{m}{}_{jk}$$

**Definition: Christoffel symbol of the 1st kind**

$$\vec{\epsilon}_n \cdot \frac{\partial \vec{\epsilon}_j}{\partial q^k} = [jk, n]$$

### Exercise: the two kinds in terms of each other

$$g^{mn}\,[jk, n] = \Gamma^{m}{}_{jk} \qquad\qquad [jk, n] = g_{mn}\,\Gamma^{m}{}_{jk}$$

Both kinds are symmetric in $j, k$, because $\dfrac{\partial \vec{\epsilon}_j}{\partial q^k} = \dfrac{\partial^2 \vec{r}}{\partial q^k \partial q^j}$ is symmetric. This is used below.

**Exercise: $\Gamma^{\mu}{}_{jk}$ is not a tensor — proof**

Go to new coordinates $q'^a(q)$. By the chain rule

$$\vec{\epsilon}'_k = \frac{\partial q^c}{\partial q'^k}\, \vec{\epsilon}_c, \qquad \vec{\epsilon}'^{\,m} = \frac{\partial q'^m}{\partial q^a}\, \vec{\epsilon}^{\,a}.$$

Then

$$\frac{\partial \vec{\epsilon}'_k}{\partial q'^j} = \frac{\partial^2 q^c}{\partial q'^j \partial q'^k}\, \vec{\epsilon}_c + \frac{\partial q^c}{\partial q'^k}\frac{\partial q^b}{\partial q'^j}\frac{\partial \vec{\epsilon}_c}{\partial b} = \frac{\partial^2 q^c}{\partial q'^j \partial q'^k}\, \vec{\epsilon}_c + \frac{\partial q^c}{\partial q'^k}\frac{\partial q^b}{\partial q'^j}\, \Gamma^{d}{}_{cb}\, \vec{\epsilon}_d.$$

Dotting with $\vec{\epsilon}'^{\,m}$ gives $\Gamma'^{m}{}_{jk} = \vec{\epsilon}'^{\,m} \cdot \partial \vec{\epsilon}'_k / \partial q'^j$:

$$\boxed{\Gamma'^{m}{}_{jk} = \frac{\partial q'^m}{\partial q^d}\frac{\partial q^c}{\partial q'^k}\frac{\partial q^b}{\partial q'^j}\,\Gamma^{d}{}_{cb} + \frac{\partial q'^m}{\partial q^c}\frac{\partial^2 q^c}{\partial q'^j \partial q'^k}}$$

A tensor would obey only the first (homogeneous) term. The second term is non-zero for any non-linear change of coordinates, so $\Gamma$ is **not a tensor**. (It vanishes only for linear transformations, e.g. Cartesian $\to$ Cartesian.)

**Exercise: ${V^j}_{;k}$ transforms as a 2nd-rank tensor — proof**

$\vec{V}$ itself is coordinate-independent, so the chain rule gives

$$\frac{\partial \vec{V}}{\partial q'^k} = \frac{\partial q^l}{\partial q'^k}\frac{\partial \vec{V}}{\partial q^l} \quad \Longrightarrow \quad {V'^j}_{;k}\,\vec{\epsilon}'_j = \frac{\partial q^l}{\partial q'^k}\,{V^i}_{;l}\,\vec{\epsilon}_i.$$

Using $\vec{\epsilon}_i = \dfrac{\partial q'^j}{\partial q^i}\vec{\epsilon}'_j$ on the right and comparing the coefficients of $\vec{\epsilon}'_j$:

$$\boxed{{V'^j}_{;k} = \frac{\partial q'^j}{\partial q^i}\frac{\partial q^l}{\partial q'^k}\,{V^i}_{;l}}$$

This is exactly the transformation law of a mixed rank-2 tensor. So although neither $\partial V^j/\partial q^k$ nor ${\Gamma^j}_{\mu k}V^\mu$ is a tensor on its own, their sum is: the inhomogeneous pieces cancel.

## 3 Christoffel Symbol in Terms of the Metric

With $g_{ij} = \vec{\epsilon}_i \cdot \vec{\epsilon}_j$ (and $g^{ij} = \vec{\epsilon}^i \cdot \vec{\epsilon}^j$):

$$\frac{\partial g_{ij}}{\partial q^k} = \frac{\partial \vec{\epsilon}_i}{\partial q^k}\cdot\vec{\epsilon}_j + \vec{\epsilon}_i \cdot \frac{\partial \vec{\epsilon}_j}{\partial q^k}$$

$$\frac{\partial g_{ij}}{\partial q^k} = [ik,j] + [jk,i] \tag{1}$$

$$\frac{\partial g_{ik}}{\partial q^j} = [ij,k] + [kj,i] \tag{2}$$

$$\frac{\partial g_{kj}}{\partial q^i} = [ki,j] + [ji,k] \tag{3}$$

Now take (1) + (2) − (3). Using the symmetry $[ab,c] = [ba,c]$, the terms $[ik,j]$ and $[ki,j]$ cancel, as do $[ij,k]$ and $[ji,k]$, while $[jk,i] + [kj,i] = 2[jk,i]$:

**Christoffel symbol of the 1st kind in terms of derivatives of $g_{ij}$**

$$\boxed{[jk,i] = \frac{1}{2}\left[\frac{\partial g_{ij}}{\partial q^k} + \frac{\partial g_{ik}}{\partial q^j} - \frac{\partial g_{kj}}{\partial q^i}\right]}$$

Raising an index with $g^{mi}$ gives the 2nd kind:

$${\Gamma^m}_{jk} = g^{mi}\,[jk,i] = \frac{1}{2}\,g^{mi}\left[\frac{\partial g_{ij}}{\partial q^k} + \frac{\partial g_{ik}}{\partial q^j} - \frac{\partial g_{kj}}{\partial q^i}\right]$$

**Exercise: do the same for $g^{ij}$**

From $\vec{\epsilon}^{\,i} \cdot \vec{\epsilon}_j = \delta^i_j$, differentiate with respect to $q^k$:

$$\frac{\partial \vec{\epsilon}^{\,i}}{\partial q^k} \cdot \vec{\epsilon}_j + \vec{\epsilon}^{\,i} \cdot \frac{\partial \vec{\epsilon}_j}{\partial q^k} = 0 \quad \Longrightarrow \quad \frac{\partial \vec{\epsilon}^{\,i}}{\partial q^k} \cdot \vec{\epsilon}_j = -\Gamma^i{}_{jk} \quad \Longrightarrow \quad \frac{\partial \vec{\epsilon}^{\,i}}{\partial q^k} = -\Gamma^i{}_{jk}\, \vec{\epsilon}^{\,j}.$$

Then with $g^{ij} = \vec{\epsilon}^{\,i} \cdot \vec{\epsilon}^{\,j}$:

$$\boxed{\frac{\partial g^{ij}}{\partial q^k} = -\,\Gamma^i{}_{mk}\, g^{mj} - \Gamma^j{}_{mk}\, g^{im}}$$

(the counterpart of $\partial g_{ij}/\partial q^k = [ik, j] + [jk, i]$).

# 4 Divergence of a Vector

$$\vec{\nabla} \cdot \vec{V}, \qquad \vec{V} = V^k \vec{\epsilon}_k$$

$$\begin{aligned} \vec{\nabla} \cdot \vec{V} &= \vec{\epsilon}^{\,j} \cdot \frac{\partial}{\partial q^j}\left(V^k \vec{\epsilon}_k\right) = \vec{\epsilon}^{\,j} \cdot \frac{\partial \vec{V}}{\partial q^j} \\ &= \vec{\epsilon}^{\,j} \cdot V^k{}_{;j}\, \vec{\epsilon}_k = V^k{}_{;j}\, \delta^j_k = V^k{}_{;k} \end{aligned}$$

$$\vec{\nabla} \cdot \vec{V} = V^k{}_{;k} = \frac{\partial V^k}{\partial q^k} + \Gamma^k{}_{kj}\, V^j, \qquad \Gamma^k{}_{kj} = \vec{\epsilon}^{\,k} \cdot \frac{\partial \vec{\epsilon}_k}{\partial q^j}$$

**Derivation of the final form (the board only showed the answer)**

**Step 1: $\Gamma^k{}_{kj} = \frac{1}{2}\, g^{km}\, \partial g_{km}/\partial q^j$.** Use $\Gamma^k{}_{kj} = g^{km}[kj, m]$ and the first-kind formula with $(j, k, i) \to (k, j, m)$:

$$\Gamma^k{}_{kj} = \frac{1}{2}\, g^{km} \left[\frac{\partial g_{mk}}{\partial q^j} + \frac{\partial g_{mj}}{\partial q^k} - \frac{\partial g_{jk}}{\partial q^m}\right]$$

In the last two terms, rename $k \leftrightarrow m$ in the second one: $g^{km}\, \partial g_{mj}/\partial q^k \to g^{mk}\, \partial g_{kj}/\partial q^m$. Because $g$ is symmetric this equals the third term, so the two cancel:

$$\boxed{\Gamma^k{}_{kj} = \frac{1}{2}\, g^{km}\, \frac{\partial g_{km}}{\partial q^j}}$$

**Step 2: Jacobi's formula.** For any invertible matrix, $\delta(\ln \det g) = \mathrm{tr}(g^{-1}\delta g)$, i.e.

$$\frac{\partial}{\partial q^j}\left[\det g\right] = (\det g)\left(g^{km}\, \frac{\partial g_{km}}{\partial q^j}\right)$$

**Step 3: combine.** Steps 1 and 2 give

$$\Gamma^k{}_{kj} = \frac{1}{2}\, \frac{1}{\det g}\, \frac{\partial (\det g)}{\partial q^j} = \frac{1}{(\det g)^{1/2}}\, \frac{\partial (\det g)^{1/2}}{\partial q^j}$$

**Step 4: the divergence.**

$$\vec{\nabla} \cdot \vec{V} = \frac{\partial V^k}{\partial q^k} + \frac{1}{(\det g)^{1/2}}\, \frac{\partial (\det g)^{1/2}}{\partial q^j}\, V^j = \frac{1}{(\det g)^{1/2}}\, \frac{\partial}{\partial q^k}\left[(\det g)^{1/2}\, V^k\right]$$

since $\dfrac{1}{\sqrt{g}}\, \dfrac{\partial}{\partial q^k}\left(\sqrt{g}\, V^k\right) = \dfrac{\partial V^k}{\partial q^k} + \dfrac{V^k}{\sqrt{g}}\, \dfrac{\partial \sqrt{g}}{\partial q^k}$.

**Divergence in general coordinates**

$$\boxed{\vec{\nabla}\cdot\vec{V} = \frac{1}{(\det g)^{1/2}}\,\frac{\partial}{\partial q^k}\Big[(\det g)^{1/2}\,V^k\Big]}$$

## 5 Laplacian

$$\nabla^2\psi = \vec{\nabla}\cdot\vec{\nabla}\psi$$

The contravariant components of $\vec{\nabla}\psi$ are $(\vec{\nabla}\psi)^k = g^{kj}\,\partial\psi/\partial q^j$ (raising the index of $(\vec{\nabla}\psi)_j = \partial\psi/\partial q^j$). Apply the divergence formula with $V^k = (\vec{\nabla}\psi)^k$:

$$\boxed{\nabla^2\psi = \frac{1}{(\det g)^{1/2}}\,\frac{\partial}{\partial q^k}\Big[(\det g)^{1/2}\,(\nabla\psi)^k\Big] = \frac{1}{(\det g)^{1/2}}\,\frac{\partial}{\partial q^k}\left[(\det g)^{1/2}\,g^{kj}\,\frac{\partial\psi}{\partial q^j}\right]}$$

**Check in spherical coordinates**

$g_{ij} = \mathrm{diag}(1,\, r^2,\, r^2\sin^2\theta)$, so $\det g = r^4\sin^2\theta$ and $\sqrt{\det g} = r^2\sin\theta$; $g^{ij} = \mathrm{diag}(1,\, 1/r^2,\, 1/(r^2\sin^2\theta))$. Then

$$\nabla^2\psi = \frac{1}{r^2\sin\theta}\left[\frac{\partial}{\partial r}\Big(r^2\sin\theta\,\frac{\partial\psi}{\partial r}\Big) + \frac{\partial}{\partial\theta}\Big(\sin\theta\,\frac{\partial\psi}{\partial\theta}\Big) + \frac{\partial}{\partial\phi}\Big(\frac{1}{\sin\theta}\,\frac{\partial\psi}{\partial\phi}\Big)\right]$$

$$= \frac{1}{r^2}\,\frac{\partial}{\partial r}\Big(r^2\frac{\partial\psi}{\partial r}\Big) + \frac{1}{r^2\sin\theta}\,\frac{\partial}{\partial\theta}\Big(\sin\theta\frac{\partial\psi}{\partial\theta}\Big) + \frac{1}{r^2\sin^2\theta}\,\frac{\partial^2\psi}{\partial\phi^2}$$

which is the standard spherical Laplacian.

## 6 Curl of a Vector

In Cartesian coordinates

$$\vec{\nabla}\times\vec{V} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ V_x & V_y & V_z \end{vmatrix}$$

**Curl in general coordinates**

$$\boxed{(\vec{\nabla}\times\vec{V})^i = \frac{1}{\sqrt{\det g}}\,[ijk]\,\frac{\partial V_k}{\partial q^j}}$$

where $[ijk]$ is the permutation symbol ($[123] = +1$, odd permutations $-1$, repeated indices $0$) and $V_k$ are the covariant components of $\vec{V}$.

**Derivation (the board only showed the answer)**

**Step 1: the curl as a derivative.** With $\vec{\nabla} = \vec{\epsilon}^{\,j}\,\partial/\partial q^j$,

$$\vec{\nabla}\times\vec{V} = \vec{\epsilon}^{\,j}\times\frac{\partial\vec{V}}{\partial q^j}.$$

Write $\vec{V} = V_k\,\vec{\epsilon}^{\,k}$. Using $\partial\vec{\epsilon}^{\,k}/\partial q^j = -\Gamma^k{}_{mj}\,\vec{\epsilon}^{\,m}$ (shown in the $g^{ij}$ exercise),

$$\frac{\partial\vec{V}}{\partial q^j} = \frac{\partial V_k}{\partial q^j}\,\vec{\epsilon}^{\,k} + V_k\,\frac{\partial\vec{\epsilon}^{\,k}}{\partial q^j} = \left(\frac{\partial V_m}{\partial q^j} - \Gamma^k{}_{mj}\,V_k\right)\vec{\epsilon}^{\,m} \equiv V_{m;j}\,\vec{\epsilon}^{\,m}.$$

So $\vec{\nabla}\times\vec{V} = V_{k;j}\,\vec{\epsilon}^{\,j}\times\vec{\epsilon}^{\,k}$.

**Step 2: the cross product of dual basis vectors.** Let $J = \vec{\epsilon}_1\cdot(\vec{\epsilon}_2\times\vec{\epsilon}_3)$ be the Jacobian $\partial(x,y,z)/\partial(q^1,q^2,q^3)$. Since $g_{ij} = \vec{\epsilon}_i\cdot\vec{\epsilon}_j$, $\det g = J^2$, i.e. $J = \sqrt{\det g}$. The dual basis is $\vec{\epsilon}^{\,1} = (\vec{\epsilon}_2\times\vec{\epsilon}_3)/J$ and cyclic. Using $(\vec{a}\times\vec{b})\times(\vec{a}\times\vec{c}) = \big[\vec{a}\cdot(\vec{b}\times\vec{c})\big]\vec{a}$:

$$\vec{\epsilon}^{\,2}\times\vec{\epsilon}^{\,3} = \frac{(\vec{\epsilon}_3\times\vec{\epsilon}_1)\times(\vec{\epsilon}_1\times\vec{\epsilon}_2)}{J^2} = \frac{J\,\vec{\epsilon}_1}{J^2} = \frac{1}{\sqrt{\det g}}\,\vec{\epsilon}_1, \qquad \text{i.e.} \qquad \vec{\epsilon}^{\,j}\times\vec{\epsilon}^{\,k} = \frac{1}{\sqrt{\det g}}\,[jkl]\,\vec{\epsilon}_l.$$

**Step 3: combine.**

$$\vec{\nabla}\times\vec{V} = \frac{1}{\sqrt{\det g}}\,[jkl]\,V_{k;j}\,\vec{\epsilon}_l \quad\Longrightarrow\quad (\vec{\nabla}\times\vec{V})^l = \frac{1}{\sqrt{\det g}}\,[ljk]\,V_{k;j}$$

(using $[jkl] = [ljk]$, a cyclic permutation).

**Step 4: the Christoffel term drops out.** $V_{k;j} = \partial V_k/\partial q^j - \Gamma^m{}_{kj}V_m$, and $\Gamma^m{}_{kj}$ is symmetric in $k,j$ while $[ljk]$ is antisymmetric, so $[ljk]\,\Gamma^m{}_{kj}V_m = 0$. Hence

$$(\vec{\nabla}\times\vec{V})^l = \frac{1}{\sqrt{\det g}}\,[ljk]\,\frac{\partial V_k}{\partial q^j}.$$

**Check:** in Cartesian coordinates $g = 1$ and $(\vec{\nabla}\times\vec{V})^1 = \partial V_3/\partial q^2 - \partial V_2/\partial q^3$, as in the determinant above.

## 7 Line Element and Scale Factors

For orthogonal curvilinear coordinates, write $\vec{\epsilon}_i = h_i\,\hat{e}_i$ with $h_i = |\vec{\epsilon}_i| = \sqrt{g_{ii}}$ (the **scale factors**). Then

$$d\vec{s} = \vec{\epsilon}_i\,dq^i = h_1\,dq_1\,\hat{e}_1 + h_2\,dq_2\,\hat{e}_2 + h_3\,dq_3\,\hat{e}_3.$$

**Consequences**

For orthogonal coordinates $g_{ij} = \mathrm{diag}(h_1^2, h_2^2, h_3^2)$, so

$$\sqrt{\det g} = h_1h_2h_3.$$

Example (spherical coordinates): $h_r = 1$, $h_\theta = r$, $h_\phi = r\sin\theta$, giving $\sqrt{\det g} = r^2\sin\theta$, consistent with the check above.